

inactivity], means extra sugar stays in your blood and, over time, affects blood vessels, leading to multi-organ diabetic complications," says Dr Sesikaran.

"Easy access to eateries and financial well-being leads to the eating of junk food. Our sedentary lifestyle and work-from-home pattern worsens the situation. Excess energy in fat leads to excess calorie accumulation that leads to diabetes," explains Dr Vidya Tickoo, endocrinologist and diabetologist, Yashoda Hospitals, Hyderabad.

Experts feel that reasons for the alarming rise in diabetes in India lie in the growing incidence of obesity. The ICMR study found that 28.6 per cent of the population would count among the obese per the BMI measure and 39.5 per cent of the population, or 351 million people, have abdominal obesity—the most frequent trigger for diabetes. "We have a genetic predisposition towards obesity and diabetes as Asians," explains Dr Kapoor. "Add to this a sedentary lifestyle and poor diet, and you have a health disaster on your hands." He emphasises that it isn't just what you eat, but also how often you eat. "If you are eating calorie-loaded food with no regulation regarding the quantity or quality of calories, you are putting pressure on your body to continuously release insulin," he adds.

Other factors also play a part in the diabetes epidemic—stress, among them. Experts say it increases cortisol levels, which causes blood sugar to spike. Lack of sleep is another contributor. In 2022, a study in the journal *Cureus* found short sleep duration was linked to insulin resistance. However, it's the lack of exercise that reduces insulin sensitivity, say experts. "Decreased physical activity due to modern technology and mechanised mobility drastically reduce energy expenditure. This leads to the weakening and loss of our muscles, which are also the storehouses of our excess sugar. Our ancestors physically exerted and burnt excess energy," explains Dr Sesikaran.

"SEDENTARY LIFESTYLE AND JUNK FOOD INCREASE RISKS. EXCESS ENERGY IN FAT LEADS TO EXCESS CALORIE ACCUMULATION, WHICH RESULTS IN DIABETES"

DR VIDYA TICKOO
Endocrinologist and diabetologist, Yashoda Hospitals, Hyderabad

EXERCISING THE BODY AND WILL

If diagnosed in time, pre-diabetics can make lifestyle changes to halt or reverse the progression of diabetes. Unfortunately, the ICMR study showed that only seven per cent of diabetics keep their BP, sugar and cholesterol in check. "Mostly, diabetes in its early stages is asymptomatic. It is picked up only if you do annual blood check-ups. It is a good practice to keep your blood sugar, cholesterol and BP in check, especially if you are overweight and have a family history of diabetes," says Dr Kapoor.

The lifestyle changes that can help prevent diabetes or reverse early signs of insulin resistance start with what we eat (see *Gastronomic Shield Against Diabetes*). "The answer is simple—going back to our ancient food habits, with less carbohydrates in the form of rice or wheat, more vegetables, salads and fruits," says Dr Tickoo. Exercise and controlling one's weight help too. Experts say these should be ingrained from childhood. "Diabetes can be reduced through awareness. People must understand that it is not just about blood sugar being high. Chronic diabetes will lead to heart problems, vision issues, reduction in quality of lifestyle. Knowing the consequences would ensure people take prevention seriously," says Dr Swamikannu M., director, internal medicine, MGM Healthcare, Chennai. For those who have diabetes or are likely to develop it, the American Diabetes Association (ADA) recommends at least 150 minutes of moderate to vigorous intensity aerobic exercise a week, spread over three days a week, and avoiding more than two consecutive days without activity.

Finally, there is consensus that health awareness is vital to reduce our country's burden of diabetes. "Family, community, schools and educational methods, public transport, walkable roads and municipal facilities, the food and beverage industry, healthcare and insurance, catering and food delivery, workplaces—all play a role in reducing diabetes," says Dr Sesikaran. Of course, with the unquestioned and unequivocal acceptance that diabetes is fast becoming a catastrophic health problem for India. A silent, predatory killer, it opens the door to multiple diseases, contributes to a high mortality rate, results in a reduced quality of life and causes great economic loss. ■

101 MILLION
No. of Indians living with diabetes, according to a survey by ICMR, the Union health ministry and the Madras Diabetes Research Foundation

30 MILLION
No. of new diabetics in the past three years in India

136 MILLION
No. of Indians in the pre-diabetes stage; nearly half of them are at risk of getting diabetes, says the survey